



SonarGuard2: Ultrasonic Face Liveness Detection Based on Adaptive Doppler Effect Feature Extraction

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Abstract

Face anti-spoofing is crucial to security of face recognition systems. Vision-based face liveness detection algorithms often fail when faced with video attacks, such as video replay. However, liveness detection based on sound waves can effectively detect such attacks by relying on the Doppler effect. In order to improve the robustness of liveness detection, we propose a novel framework, named SonarGuard2, adaptively selecting the ultrasonic signals and analyse the Doppler effect. Specifically, we introduce acoustic echo cancellation to filter to get the Doppler effect features, then utilize complex convolutional neural networks to enhance the ability to model the Doppler effect features. Furthermore, we propose a novel selection strategy to determine the usability of ultrasonic signals on mobile devices. The performance and visualization on the collected data demonstrate the effectiveness of our framework.

Index Terms: liveness detection, complex convolutional neural network, doppler effect, acoustic echo cancellation

1. Introduction

With the proliferation of mobile devices, the demand for mobile payments is also increasing. To ensure the security of these payments, liveness detection using image information is a commonly employed method. However, vision-based liveness detection is vulnerable to video replay attacks [1], as these attacks utilize real images. It is challenging to determine whether an image represents a live person or an attack based on image information alone. The micro-Doppler effect [2] of signals reflected from moving objects can effectively resist video replay attacks. Incorporating Doppler effect information of object motion can enhance the robustness of a liveness detection system. For widely used mobile devices such as smartphones, sound waves are the most easily accessible signals. To ensure a good user experience, the emitted signal should be imperceptible. Therefore, using ultrasonic signals as the emitted signal is a reasonable choice.

However, due to the varying positions of the signal emission and reception modules, as well as differences in the built-in signal processing algorithms of mobile devices, some devices may be unable to emit ultrasonic signals, or their reception modules may apply noise reduction processing that distorts the ultrasonic signals. In practical applications, it is necessary to have a method to determine whether the received ultrasonic signals are usable. We propose two criteria based on signal integrity and signal distortion to make this assessment. Our work builds upon SonarGuard [3], which uses ultrasonic signals to extract the Doppler effect spectrum for liveness detection. In practical applications, we found that SonarGuard, which relies on the Doppler effect of lip movements, has stringent requirements for

the signal emission angle. Many mobile phones do not meet these requirements, limiting its application range. To expand the practical application range of the algorithm, we shifted from lip movements to forward and backward movements of the face to obtain a more pronounced Doppler effect spectrum, thereby reducing dependence on the signal emission angle. To make SonarGuard adaptable to facial movements, we made several adjustments. Based on SonarGuard's ultrasonic-only network, we replaced the Doppler effect feature extraction with an adaptive filtering method. We use complex convolutional neural networks [4] to directly enhance the signal and detect liveness. These adjustments enable our solution to be fully data-driven, achieving adaptive Doppler effect feature extraction.

We collected a basic dataset under normal usage conditions, as well as a challenge dataset that includes hand movement interference. Our method achieves an average error rate of 0.59% on the basic dataset and 13.01% on the challenge dataset.

2. Related Work

To the best of our knowledge, there has been limited research on using ultrasound for liveness detection over the past few decades. Most of the existing work has focused on using ultrasound for gesture recognition on mobile devices [5,6,7,8]. Since vision-based liveness detection cannot effectively resist replay attacks that use real liveness data, ultrasound, which can easily counter such attacks, has recently started to gain attention for liveness detection.

References [9,10,11] utilize ultrasonic reflection signals to determine liveness through the Doppler effect of lip movements. This method relies heavily on numerous preset signal processing thresholds, making it suitable only for specific scenarios. Reference [12] employs two microphones to detect face liveness, but hardware differences between microphones on different devices can significantly impact performance. Reference [13] proposes a face verification scheme that combines audio and image. It designs a special transmission signal; however, since the duration of this signal is very short and its frequency changes over time, the liveness detection system cannot obtain a complete and clear Doppler effect.

Our previous work, SonarGuard, was the first to explore the use of an attention structure to fuse image and ultrasound information for face liveness detection [3]. Compared to vision-based liveness detection, it can resist video replay attacks. However, we identified several issues in practical applications. Firstly, the Doppler effect of lip movement is not pronounced. For a clear Doppler effect, the signal needs to be emitted in a specific direction, a requirement that most devices cannot meet. Secondly, after the ultrasound signal is demodulated and processed, it passes through a notch filter and a low-pass filter. The

threshold for the low-pass filter, which is set based on experience, and the filtering of the direct current (DC) component can result in the loss of the Doppler effect. This loss can negatively impact the accuracy of liveness detection. Thirdly, SonarGuard's beamforming method [14] assumes a set of ultrasonic signals reflected by moving objects through the maximum value of the signal difference after beamforming. However, due to the varying distance between the user and the verification device and differences in the user's movement posture, these variables can lead to inaccurate ToF estimates. When the ToF estimation is incorrect, the accuracy of liveness identification drops rapidly.

3. Method

To address the issues encountered by SonarGuard in practical applications, we made several key modifications. First, we shifted from detecting lip movements to detecting the forward and backward movements of the face. This change leverages larger movement amplitudes, thereby reducing the directional requirements for the transmitted signal. To solve the problem of incomplete Doppler effect extraction, we employed an acoustic echo cancellation method [15] to filter out the transmitted ultrasound signals, thereby retaining the Doppler effect portion. This approach adaptively extracts complete Doppler effect information. To address the issue of inaccurate Time of Flight (ToF) estimation, we utilized a complex convolutional neural network. The input signal, after undergoing Short-Time Fourier Transform (STFT), is directly fed into the complex convolutional neural network. Since complex numbers contain information about both phase and energy, and the phase represents the spatial position of the object [14], the complex convolutional neural network can adaptively learn the most suitable ultrasonic signal features. Our ultrasonic liveness detection system is divided into three main modules: the ultrasonic availability judgment module, the acoustic echo cancellation module, and the liveness detection module. The overall structure is shown in Figure 1. The details of each module are described below.

3.1. Ultrasonic availability judgment module

The system first emits three ultrasonic carriers [16,17,18] with a frequency interval of 100 Hz. In our system, the demodulated signals of these carriers are used to determine the ultrasonic availability of the device. The ultrasonic emission signal can be described as

$$s(t) = \sum_{i=1}^N (A \cos 2\pi f_i t + A \sin 2\pi f_i t) \quad (1)$$

where f denotes the signal frequency, N denotes the number of carrier we transmit, i, t denote the frequency and time index, respectively. The transmitted ultrasonic signal follows multiple propagation paths, including reflections from various objects. The received signal can be described as

$$R(t) = \sum_{k=1}^M \sum_{i=1}^N (A_k^i(t) \cos(2\pi f_i t + \theta_k^i(t)) + A_k^i(t) \sin(2\pi f_i t + \theta_k^i(t))) \quad (2)$$

where M denotes the number of ultrasonic propagation paths, $A_k^i(t)$ represents the energy attenuation of the i -th carrier on the k -th propagation path, and $\theta_k^i(t)$ represents the phase change of the i -th carrier on the k -th propagation path.

We perform IQ demodulation on the received signal using each carrier. After low-pass filtering, the demodulated signal

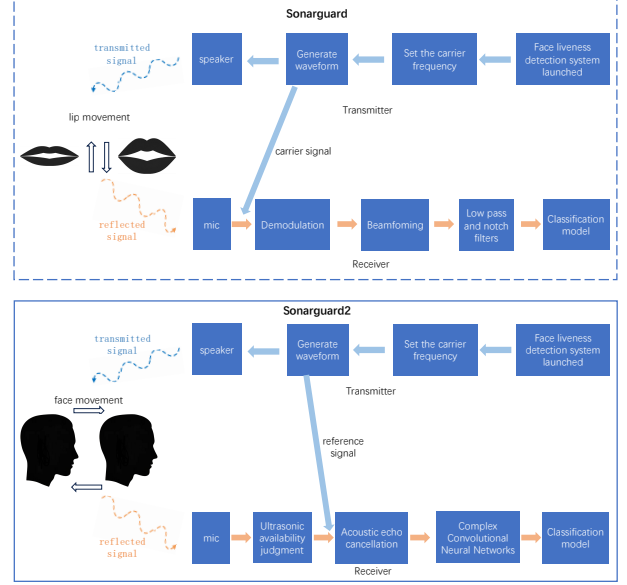


Figure 1: The overall framework of SonarGuard2 and its comparison with SonarGuard.

can be described as

$$\begin{aligned} I_j(t) &= \sum_{k=1}^M \sum_{i=1}^N (A_k^i(t) \cos(2\pi \Delta f_{ij} t + \theta_k^i(t)) \\ &\quad + A_k^i(t) \sin(2\pi \Delta f_{ij} t + \theta_k^i(t))) \\ Q_j(t) &= \sum_{k=1}^M \sum_{i=1}^N (A_k^i(t) \cos(2\pi \Delta f_{ij} t + \theta_k^i(t)) \\ &\quad - A_k^i(t) \sin(2\pi \Delta f_{ij} t + \theta_k^i(t))) \end{aligned} \quad (3)$$

where j denotes the index of the carrier used for demodulation, Δf_{ij} denotes $f_i - f_j$. $I_j(t)$ and $Q_j(t)$ can be simplified by adding and subtracting, which can be described as

$$\begin{aligned} I_j(t) + Q_j(t) &= \sum_{k=1}^M \sum_{i=1}^N A_k^i(t) \cos(2\pi \Delta f_{ij} t + \theta_k^i(t)) \\ I_j(t) - Q_j(t) &= \sum_{k=1}^M \sum_{i=1}^N A_k^i(t) \sin(2\pi \Delta f_{ij} t + \theta_k^i(t)) \end{aligned} \quad (4)$$

Since the frequency interval of the carrier is 100 Hz, Δf_{ij} is equal to multiples of 100 Hz. When using the lowest carrier frequency for demodulation, the demodulated signal consists of DC, 100 Hz, and 200 Hz components, corresponding to three different Δf_{ij} . The phase and amplitude attenuation of $I - Q$ and $I + Q$ at 100 Hz and 200 Hz are the same, so their energy at these frequencies should also be the same. However, at 0 Hz, the energy of $I + Q$ and $I - Q$ differs because $\theta_k^i(t)$ contains a static phase component. The phase of the static part does not change with time, and the results of the cosine and sine calculations on the phase are different. This conclusion helps us derive a criterion for signal integrity, if the $I + Q$ and $I - Q$ signals demodulated by the lowest frequency carrier have different energies at 100 Hz or 200 Hz, then the signal is incomplete and is judged to be an invalid ultrasonic signal.

To determine whether the received signal is distorted, we use the cross-correlation coefficient between the transmitted signal and the received signal. If the cross-correlation coefficient exceeds the set threshold, the signal can be considered undistorted. Based on the above analysis, we have established two criteria for signal availability: the signal integrity criterion and the signal distortion criterion.

3.2. Acoustic echo cancellation module

We use a common acoustic echo cancellation technique [19] to extract Doppler effect features. The Doppler effect can cause changes in the frequency of the reflected signal from a moving object. To extract this feature, our previous work involved using the demodulated signal to remove the DC component and then applying a low-pass filter to retain the spectrum below 50 Hz. The collected data show that these low-frequency spectra preserve most of the Doppler effect.

However, when the range of motion of the object is large, such as the movement of an entire face, these filters cannot effectively extract Doppler effect features. To extract Doppler features in different scenarios, we replace demodulation and feature extraction with acoustic echo cancellation. We use three ultrasonic carriers as reference signals and the received signal as the input signal. After acoustic echo cancellation, the ultrasound component of the received signal is eliminated, leaving only the Doppler effect features. Acoustic echo cancellation adjusts parameters adaptively according to the data, making it suitable for Doppler effect feature extraction in various scenarios. Since there is no low-pass filtering or notch filtering operation, complete Doppler effect features are retained.

3.3. Liveness detection module

SonarGuard uses the estimated Time of Flight (ToF) to determine the phase difference of each carrier [3], and then enhances the signal of the moving object by adjusting the phase delay of each carrier. However, ToF estimation often fails when the motion range is large. To avoid ToF estimation errors in practical use, we employ a complex convolutional neural network to combine ToF estimation with liveness detection. Complex signals carry both energy and phase information, with phase information being related to the spatial position of the object. By using the complex signal after Short-Time Fourier Transform (STFT), we can process multiple carrier signals with different phase delays and optimize the parameters of the complex convolutional neural network and classification network through classification loss.

The complex signal is processed by the complex convolutional neural network and then sent to the classification model. The classification model utilizes LSTM [20,21] and the encoder layer of the transformer [22] to capture both local and global information, employing focal loss [23] as the overall loss function. We use attention pooling [24] to map each sample with different time frames into a fixed-dimension vector, which is then used for classification.

4. Experiments

4.1. Data

Our datasets are divided into two categories: basic datasets and challenge datasets. These datasets were collected using ten different models of mobile phones. The basic dataset contains 1,620 samples, including 809 living samples and 811 attack samples. The challenge dataset contains 530 samples, including 258 living samples and 272 attack samples. By employing data augmentation techniques, such as randomly extracting sample segments during object movement, we increased the size of the dataset tenfold. The living samples fall into two types: the first involves the face moving away from the phone, then closer, and then away again; the second type involves the opposite movement pattern. In the basic dataset, the attack samples consist of

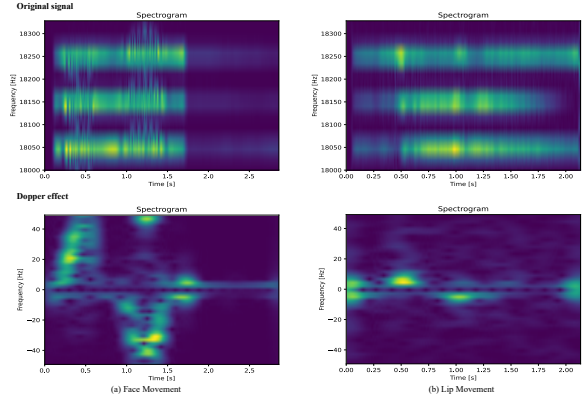


Figure 2: Comparison of Doppler effect spectrum of face and lip motion, the carrier frequencies are 18050hz, 18150hz, 18250hz.

face movements played in a video, mimicking the same movement patterns as the living samples. In the challenge dataset, the attack samples added interference from human hand movements during video replay. We randomly divided the datasets into training and test sets in a ratio of 4:1. We filter all samples through the ultrasonic availability judgment module before training and testing.

4.2. Implementation details

Our model consists of a complex convolutional layer, two LSTM layers, and two transformer encoder layers. The kernel size of the complex convolution is 3, with 16 output channels. The LSTM has 64 hidden units, and the transformer’s attention dimension is 128. The structure has a total of 3.1 million parameters. The models were developed using PyTorch [25]. We used the adam optimizer for training, with an initial learning rate of $1e-4$, and trained the model for 100 epochs.

4.3. Baseline and Comparison Models

We use SonarGuard’s ultrasonic-only network for comparison. To further compare our model with commonly used audio classification models, we also conducted experiments using the Convolutional Recurrent Neural Network (CRNN) [26]. The parameters of the models we compared are similar to those of our proposed model. To assess the effectiveness of our proposed ultrasonic availability judgment module, we randomly selected 300 samples and manually labeled the samples, dividing them into three categories: lost ultrasonic signal, impaired ultrasonic signal, and normal ultrasonic signal.

4.4. Evaluation Metrics

We used the same evaluation metrics as SonarGuard: Attack Presentation Classification Error Rate (APCER), Bona Fide Presentation Classification Error Rate (BPCER), and Average Classification Error Rate (ACER) [27].

5. Results

5.1. Comparison of the Doppler effect

Figure 2 illustrates the comparison of the Doppler effect spectrum between lip movement and face movement. It is evident that the frequency variation in face movement is greater than

Table 1: Classification error rates (%) of different methods on our basic dataset and challenge dataset.

Testset	Method	APCER	BPCER	ACER
challenge	CRNN	38.46	40.54	39.50
	Sonarguard	37.50	25.00	31.25
	Sonarguard2	19.35	6.67	13.01
basic	CRNN	4.48	27.45	15.96
	Sonarguard	9.09	19.42	14.25
	Sonarguard2	0.00	1.18	0.59

Table 2: Ablation study based on Sonarguard2

Testset	Sonarguard2	APCER	BPCER	ACER
challenge	w/o cconv	17.24	21.28	19.26
	w/o aec	21.05	15.79	18.42
basic	w/o cconv	2.20	7.69	4.95
	w/o aec	2.35	2.38	2.37

that in lip movement, making the Doppler effect more pronounced. This also implies reduced hardware requirements for mobile devices.

5.2. Overall Results

Table 1 presents the comparison of classification error rates across different methods. Sonarguard2 has achieved the best results on our datasets. In contrast, the audio classification model CRNN exhibits the highest classification error rate, primarily because it uses mel features, which have low resolution at high frequencies. The Sonarguard shows a high classification error rate due to its beamforming not being well-suited for large motion ranges, such as face movement. The Time of Flight (TOF) estimated by Sonarguard is inaccurate for objects with large motion ranges, and its beamforming negatively affects accuracy. In our experiments, we found that the classification error rate of Sonarguard without beamforming was reduced to some extent on the basic dataset. However, the classification error rate remained essentially unchanged on the challenge dataset. This is because the incomplete Doppler effect caused by filtering prevents Sonarguard from effectively distinguishing between different moving objects.

5.3. Ablation Study

Table 2 presents the effects of aec and cconv (complex convolution) on Sonarguard2. It is evident that cconv has a greater impact on the classification error rate compared to aec. This is because the parameters of cconv are adjusted based on the entire training dataset, whereas the parameters of aec are optimized using only a single sample. Consequently, cconv has better generalization ability. And cconv also has the ability of feature enhancement, which can learn the most suitable feature.

5.4. Evaluation of ultrasonic availability

Table 3 presents the number of invalid data detected by ultrasonic availability judgment module. It can be observed that the module can detect all samples with lost ultrasonic signals, does not misjudge normal samples, and maintains high detection accuracy for samples with impaired ultrasonic signals.

Table 3: Results of ultrasonic availability judgment module.

Ultrasonic signal	Number	Invalid data detected
Lost	55	55
Impaired	67	65
Normal	178	0

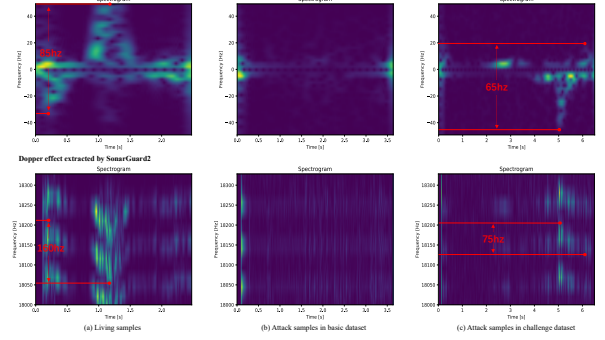


Figure 3: Doppler effect spectrum of different samples, the carrier frequencies are 18050hz, 18150hz, 18250hz.

5.5. Doppler effect spectrum analysis

Figure 3 presents the Doppler effect spectrum after processing by Sonarguard and after Sonarguard2. The upper and lower spectrum in each column represent the Doppler effect of the same sample processed by different methods. Sonarguard used notch filters to remove the DC component due to the strong DC energy that can interfere with the Doppler effect. In contrast, Sonarguard2 fully preserves the Doppler effect across all frequency bands. In the challenge dataset, the Doppler effect of attack samples processed by Sonarguard2 is noticeably different from that of living samples. Specifically, the frequency variation in living samples (160hz) is greater than that in attack samples (75hz). The frequency variation of sonarguard1 is smaller than that of sonarguard2. Since Sonarguard sets 50 Hz as the threshold for its low-pass filter, it can only retain Doppler effect frequency changes up to 50 Hz. In contrast, Sonarguard2 adaptively retains the Doppler effect, thereby imposing no limit on the frequency change range of the Doppler effect. This indicates that the Doppler effect features we extracted are both complete and distinguishable.

6. Conclusion and Future Work

We investigated the application of ultrasonic signals in face liveness detection. For various types of mobile devices, we proposed criteria for assessing the availability of ultrasonic signals. To address the issue of reduced generalization caused by empirically preset filtering thresholds in Doppler effect feature extraction, we employed Acoustic Echo Cancellation (AEC) to adaptively filter the transmitted signal, thereby retaining the complete Doppler effect. To tackle the problem of inaccurate estimation of phase difference between carriers, we utilized a complex convolutional neural network to directly process complex signals containing both energy and phase information. As a result, our algorithm achieved the best performance. In the future, we plan to integrate image information to further enhance the accuracy of liveness detection.

7. References

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